

Tutorial 3 Questions and Solutions

Introduction to Programming

Q1.

Usually, when you buy something, you're asked whether your credit card number, phone number, or answer to your most secret question is still correct. However, since someone could look over your shoulder, you don't want that shown on your screen. Instead, we mask it.

Your task is to write a function **maskify()**, which changes all but the last four characters into '#'.

Example

```
maskify("4556364607935616") => "#####5616"
maskify("64607935616") => "#####5616"
maskify("1") => "1"
maskify("") => ""
```

Possible solution:

SOL1

```
def maskify(cc):
    str = ""
    if len(cc)>4:
        for i in cc[:len(cc)-4]:
            str = str+'#'
        for i in cc[len(cc)-4:len(cc)]:
            str = str+i
    else:
        str = cc
    return str
```

Also adding other possible solutions

SOL2:

```
def maskify(cc):
    return "#"*(len(cc)-4) + cc[-4:]
```

SOL3:

```
def maskify(cc):
    l = len(cc)
    if l <= 4: return cc
    return (l - 4) * '#' + cc[-4:]
```

SOL4:

def maskify(cc):

```
word = list(cc)
#loop through the list except the last 4 index's this will also prevent
#the loop from running for anything less than 5 index's long
for i in range(len(word) - 4):
    word[i] = '#'
# join and return the list
return "".join(word)
pass
```

SOL5

```
def maskify(cc):
    newstring = ""
    for character in cc[0:-4]:
        newstring += '#'
    for number in cc[-4:]:
        newstring += number
    return newstring
```

Q2:

Deoxyribonucleic acid (**DNA**) is a chemical found in the nucleus of cells and carries the "instructions" for the development and functioning of living organisms.

In DNA strings, symbols "A" and "T" are complements of each other, as "C" and "G". You take *input* a **string**, as one side of the DNA; you need to return the other **complementary** side.

DNA strand is never empty or there is no DNA at all.(i.e, if the input is an empty string, "", then simply print out the empty string)

Also, if the input DNA string contains some wrongly encoded character, it prints out "Wrong Encoding".

Example (input -> output)

"ATTGC" → "TAACG"

"GTAT" → "CATA"

"ATGC" → "TACG"

"GTAT" → "CATA"

"AAAA" → "TTTT"

" " → " " ## empty string

```
dna = input()
comp_dna = ""
Encoding = True
## initialized the complementary part of the dna with
## empty string
for x in dna:
    if x == " ":
        ## That is the dna is empty
        print("")
        ##no complementary side as no dna exists
```

```
        break
    else:
        if x == 'A':
            comp_dna = comp_dna + 'T'
        elif x == 'T':
            comp_dna = comp_dna + 'A'
        elif x == 'G':
            comp_dna = comp_dna + 'C'
        elif x == 'C':
            comp_dna = comp_dna + 'G'
        else:
            print("Wrong Encoding!!")
            Encoding = False
            break

if (Encoding == True):
    ## or simply can use `if(Encoding)`
    print(comp_dna)
```